

K3 audit — Engine + Bulk-color comprehensive cluster K-audit. Sit with the Saturday-cumulative work as ONE structural unit: Lyra engine v0.1-v0.4 + Elie engine v0.2 + engine support v0.2 (Toys 3612-3626) + Lyra bulk-color v0.1-v0.6. Formal K-audit with per-component structural state, full closure-condition matrix (C1-C7), multi-week roadmap, cross-CI handoffs, tier dispositions, v0.7+ pre-staging. **CONDITIONAL PASS** (substantive advance; closure verification multi-week).

Keeper

2026-05-30 Saturday afternoon

K3 Audit — Engine + Bulk-color comprehensive cluster

0. Scope and framing

K3 is the cluster K-audit treating the substrate's dynamics engine + the bulk-color SU(3) emergence frontier as ONE structural unit. The Saturday batch produced ~10 substantive component docs across Lyra + Elie; this K-audit synthesizes them, identifies the load-bearing structural claims, fixes the closure-condition matrix for v0.7+ work, and gives the cross-CI handoff specifications.

Components in scope: - Engine layer (substrate Hall algebra + Hopf structure): - Lyra engine v0.1-v0.4 (Hall multiplication + Green coproduct + R-matrix + negative part) - Elie engine v0.2 + audit support v0.2 (15 toys 3612-3626 consolidating Drinfeld double, CPT, bulk-color algebraic §6/§7 candidates) - Bulk-color frontier (SU(3) emergence on D_{IV}^5): - v0.1 mechanism mapping (3 candidate directions) - v0.2 SO(3) sub-vector counting layer (CANDIDATE) - v0.3 team integration (Family 4 FAVORED counting-not-symmetry) - v0.4 route commitment (C+D unified) - v0.5 Toeplitz setup + count match (RETIRED in favor of v0.6) - v0.6 Cartan-Weyl basis correction ($8 = 3 T_a + 3 T_a^\dagger + 2 K\text{-Cartan}$, structurally aligned with $\mathfrak{su}(3)$)

Verdict: **CONDITIONAL PASS** at CANDIDATE FRAMEWORK with explicit closure-condition matrix.

1. Per-component structural state

1.1 Engine layer — substrate Hall algebra at $q=2$

Subsystem	Component	State
Multiplication	Lyra engine v0.1: Hall product structure constants = substrate primaries	RIGOROUS (E0/3597 short Serre $[2]_2 = 3 = N_c + E9/3610$ long Serre $[3]_4 = 21 = N_c \cdot g$)
Coproduct (decay)	Lyra engine v0.2: Green coproduct decomposition; β -decay (E3/3601) as worked example	RIGOROUS as algebra; physics ID modulo keystone

Subsystem	Component	State
R-matrix (scattering)	Lyra engine v0.3: Drinfeld double R-matrix; Yang-Baxter	FRAMEWORK (structure exists; amplitude extraction multi-week)
Negative part (antimatter)	Lyra engine v0.4: F-generators; bar involution = CPT	RIGOROUS as algebra; Elie A6/3617 verifies Drinfeld pairing numerators encode $N_c + N_c \cdot n_C$
Cartan grading (conservation)	engine v0.2 grading	RIGOROUS; Q, B, L automatically preserved via grading on Hopf algebra
Affine pin	Lyra v0.1 + Keeper refinement	PINNED: $B_2^{(1)}$ marks (1,1,2), $\delta = \alpha_0 + \alpha_1 + 2\alpha_2$, $h=4$
Langlands duality (NEW Saturday)	Lyra B3 session	STRUCTURAL REFRAME: $B_2^{(1)} \leftrightarrow C_2^{(1)}$ IS Langlands duality at affine level

Engine v0.2 $\rightarrow$ v0.3 candidates (Elie audit support v0.2): - §6 Drinfeld double + CPT (Toy 3617): F-mirror exact; Drinfeld pairing carries N_c , $N_c \cdot n_C$; $\omega/\sigma/W_0 \rightarrow C/T/P$. STRUCTURAL candidate. - §7 Bulk-color algebraic (Toy 3620): $SO(5) \supset SO(3) \times SO(2)$ decomposes $5 = N_c + \text{rank}$. STRUCTURAL candidate. - K1.2 + K1.4 audit anchors pre-reserved for when Elie files the engine v0.3 consolidation doc.

1.2 Bulk-color frontier — sub-gate (a) counting layer

State: ADDRESSED structurally.

Component	State
Lyra v0.2 $SO(3)$ sub-vector: 5 of $SO(5) \downarrow SO(3) \times SO(2) = (3,0) + (1,\pm 1)$; 3-dim $SO(3)$ sub-vector hosts color labels in bulk holomorphic tangent	CANDIDATE STRUCTURAL (Lie-theoretically rigorous)
Elie A1 (Toy 3612) Cartan decomposition $\mathfrak{so}(5,2) = \mathfrak{k} \oplus \mathfrak{p}$, $\dim 21 = 11+10$	RIGOROUS
Two routes for direct $SU(3)$ subgroup CLOSED: $SU(3)$ not in $\mathfrak{k}$ ($B_2 \neq A_2$); $SU(3)$ not in $\mathfrak{p}$ ($[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$)	RIGOROUS structural narrowing
Cross-confirmation via engine v0.3 §7 candidate ($5 = N_c + \text{rank}$)	STRUCTURAL cross-anchor

1.3 Bulk-color frontier — sub-gate (b) gauge-phenomenology layer

State: CANDIDATE COUNT MATCH at IDENTIFIED tier, structurally aligned with $\mathfrak{su}(3)$ Cartan-Weyl basis per v0.6 correction; closure verification multi-week.

Component	State
Toeplitz algebra $T(H^2(S))$ non-commutative even when boundary multiplication abelian	RECALLED (Upmeyer/Vasilevski) — C1 sourcing condition outstanding
Hardy-projection-induced compact commutators give non-abelian content	RIGOROUS structurally (short exact sequence $0 \rightarrow K \rightarrow T(H^2(S)) \rightarrow C(S) \rightarrow 0$)
Cartan-Weyl 8-decomposition (v0.6 correction): $8 = 3 T_a + 3 T_a^{\wedge \dagger} + 2 K$ -Cartan	CANDIDATE STRUCTURALLY ALIGNED with $\mathfrak{su}(3)$

Component	State
v0.5 alternative decomposition ($3 T_a + 3 [T_a, T_b] + 2 K\text{-Cartan}$)	RETIRED — structurally mismatched for $su(3)$; $[T_a, T_b]$ are DERIVED compact operators, not independent generators
$su(3)$ closure verification: explicit commutation relations	OPEN multi-week (C4)
QCD low-energy phenomenology (confinement, asymptotic freedom, running coupling)	OPEN multi-week

2. Closure-condition matrix (consolidated)

Five conditions carry forward to v0.7+:

#	Condition	Owner	Estimated horizon
C1	Toeplitz-source citation (Upmeier <i>Toeplitz Operators and Index Theory in Several Complex Variables</i> or Vasilevski) VERIFIED-CITED with theorem/page for the load-bearing claim “ $T(H^2(S))$ non-commutative even when boundary multiplication abelian” applied to Shilov boundary of D_{IV}^5 specifically (Cal #33 STANDING)	Lyra	days
C2	AdS/CFT analogy demoted from “substantively similar” to “evocative parallel, not structural argument”	Lyra	minutes (wording fix in v0.7+)
C3	$SO(3)$ -decomposition mismatch	CLOSED via v0.6 Cartan-Weyl basis ($3 T_a + 3 T_a^{\dagger} + 2 \text{ Cartan}$); no further action needed	n/a
C4	Explicit Toeplitz commutator computation: (a) $[T_a, T_a^{\dagger}] = \alpha(H_?) \leftrightarrow$ substrate-natural Cartan element; (b) $[T_a, T_b] = N_{\{\alpha\beta\}} T_{\{\alpha+\beta\}} \leftrightarrow$ root-sum structure	Elie	weeks-multi-week

#	Condition	Owner	Estimated horizon
C5	Hermiticity + semisimplicity + tracelessness verification: does the algebra $\{T_a, T_a^\dagger, K\text{-Cartan}\}$ closure give $\mathfrak{su}(3)$ (vs $\mathfrak{u}(3)$ or other 8-dim)	Elie	weeks
C6	Identification of $\mathrm{SU}(3)$ Cartan eigenvalues with quark color quantum numbers	Lyra	weeks (post C4+C5)
C7	QCD low-energy phenomenology emergence (confinement, asymptotic freedom, running coupling)	Lyra + Elie	multi-week to month+

3. Multi-week roadmap (sequenced)

Phase 1 (days-to-weeks) — close C1+C2 + start C4: - Lyra v0.7 absorbs C1+C2 (Toeplitz sourcing pin + AdS/CFT demotion) - Elie starts explicit Toeplitz commutator computation $[T_a, T_a^\dagger]$ and $[T_a, T_b]$ on $H^2(S)$

Phase 2 (weeks) — C4+C5: - Elie computes commutator structure constants - Lyra absorbs into v0.8: does the algebra close $\mathfrak{su}(3)$? - If YES: Phase 3; if NO: structural rework or retreat to count-aligned-OPEN disposition

Phase 3 (weeks-multi-week) — C6: - Lyra + Elie identify $\mathrm{SU}(3)$ Cartan eigenvalues $\leftrightarrow$ quark color quantum numbers - Engine v0.4 absorbs the closure result

Phase 4 (multi-week to month+) — C7: - QCD low-energy phenomenology derivation: confinement mechanism via Toeplitz/compact-operator structure on Shilov boundary; asymptotic freedom; running coupling - This is full bulk-color $\mathrm{SU}(3)$ closure of gate (b)

At each phase: K-audit refresh; Cal cold-read; Honest-State Ledger update.

4. Cross-CI handoff specifications

To Lyra

- v0.7 bulk-color: absorb C1 (Toeplitz-source cite) + C2 (AdS/CFT demotion). Mechanical absorption ~30 min.
- Engine v0.3 §3 R-matrix $\leftrightarrow$ Toeplitz distinction (per my Vol 16 §4 N1): “R-matrix-induced Toeplitz commutators” reframed to acknowledge they’re distinct mechanisms.
- Engine v0.3 consolidation doc: pending Elie filing of the actual doc with §6+§7 sections.
- Quasi-Eigentone v0.3: disambiguate Resolution B reading (1 over-determination vs 2 structurally-independent $C_2=6$ instances per my v0.2 tier-gate two-structures-burden flag).

To Elie

- C4 explicit Toeplitz commutator computation — THE bulk-color closure work; multi-week starting now
- C5 algebra-closure verification ($\mathfrak{su}(3)$ vs $\mathfrak{u}(3)$)
- Engine v0.3 consolidation doc filing (with §6 Drinfeld/CPT + §7 bulk-color algebraic) — K1.2 + K1.4 audit anchors reserved
- Bulk K-type radial tower for L4 v0.2 mass spectrum (parallel track, separate from bulk-color closure)

To Grace

- Catalog absorption of v0.6 Cartan-Weyl $3+3+2 = 3 T_a + 3 T_a^\dagger + 2 K$ -Cartan (G13 8-gluon scaffold update)
- Per-cell stability tagging in Periodic Table v0.8 accounting for v0.6 corrected decomposition
- “Live tests” column for F1-F5 / T1-T5 falsifiers + bulk-color closure milestone tracking

To Cal

- Cold-read v0.6 bulk-color verify Cartan-Weyl basis structural alignment; flag any remaining issues. Especially: is the $T_a^\dagger$ assignment the standard Hermitian conjugate or substrate-specific structure?
- Cold-read engine v0.3 consolidation doc when Elie files ($K1 \cdot 2 + K1 \cdot 4$ cross-anchor verification).
- Pattern-watch toward Calibration #35 candidate (three-instance threshold).

To Casey

- This K-audit ratifies bulk-color closure as a MULTI-WEEK structural program with explicit closure-condition matrix C1-C7.
- The Saturday advance (v0.4 commitment $\rightarrow$ v0.5 count match $\rightarrow$ v0.6 Cartan-Weyl correction) is substantive; closure verification is the multi-week burden.
- SP-30 outreach pre-drafts (Task #330) are the most productive parallel-track work to advance Substrate Engineering during the bulk-color multi-week closure.

5. Tier dispositions (consolidated)

Subsystem	Tier	Notes
Hall multiplication / coproduct / R-matrix / negative part	DERIVED-modulo-keystone at algebraic level	Engine 4-piece Hopf structure RIGOROUS as algebra; physics ID rides on dictionary keystone
Mersenne-coded structure constants ($\{N_c, N_c \cdot n_C, N_c \cdot g, C_2\}$ at $q=2$)	DERIVED	Mersenne-Hall correspondence confirmed
Cartan grading conservation (Q, B, L)	DERIVED	grading on Hopf algebra automatic
Affine pin $B_2^{(1)} +$ Langlands-duality framing	PINNED + STRUCTURAL REFRAME	Saturday B3 session
Engine v0.3 §6 (Drinfeld + CPT)	STRUCTURAL candidate	pre-reserved $K1 \cdot 2$ anchor
Engine v0.3 §7 (bulk-color algebraic)	STRUCTURAL candidate	pre-reserved $K1 \cdot 4$ anchor; cross-confirms bulk-color (a) counting layer
Bulk-color (a) counting layer	CANDIDATE-ADDRESSED	SO(3) sub-vector geometric; cross-confirmed by §7
Bulk-color (b) gauge-phenomenology layer	CANDIDATE COUNT MATCH at IDENTIFIED	v0.6 Cartan-Weyl alignment; closure verification multi-week
QCD low-energy emergence (confinement, asymptotic freedom)	OPEN multi-week	C7

6. K3 verdict + audit-chain governance

K3 CONDITIONAL PASS at CANDIDATE FRAMEWORK level.

Conditions: C1+C2 carry from v0.4; C4+C5+C6+C7 are the multi-week closure burdens. C3 closed via v0.6 correction.

Per audit-chain governance (Casey delegation 2026-05-17): D-tier promotion would require Cal cold-read PASS on v0.6 Cartan-Weyl alignment AND explicit closure verification (C4+C5). Current state: pre-D-tier candidate; promotion to RATIFIED-status awaits multi-week verification.

Headline-cap-conditionality (Calibration #34 STANDING): every external mention of “bulk-color SU(3) emergence” must carry “CANDIDATE COUNT MATCH at Cartan-Weyl basis alignment; closure verification multi-week.” The bare “8-gluon emergence” framing is too strong; the conditional cap travels with the headline.

Saturday’s substantive advance: v0.4 (commitment) → v0.5 (count match, structurally wrong) → v0.6 (Cartan-Weyl correction, structurally aligned) is the kind of sustained, self-correcting progress the team’s R3 sustainable cadence + Cal #27 STANDING discipline produces in real time. The bulk-color frontier moved one substantive step closer to closure today.

7. K3 ↔ K1 ↔ K2 audit-chain context

K#	Subject	State
K1	Engine v0.2 (Lyra L1-L4)	CONDITIONAL PASS (Friday) with 3 findings; substantively cleared via Elie 3609/3610 + Lyra v0.2
K2	Dictionary first wave (L1-L4 + lepton row)	CONDITIONAL PASS (Friday) with 3 findings; absorbed into Lyra v0.2 sweep
K3	Engine + Bulk-color comprehensive cluster (Saturday)	CONDITIONAL PASS (this audit) with 6 carrying conditions (C1+C2 + C4-C7); C3 closed via v0.6

The K-audit chain through Saturday late-morning: K1 → K2 → K3, each at CONDITIONAL PASS with explicit conditions carrying to the next version cycle.

8. Closing

The Saturday cumulative work on engine + bulk-color is substantial and structurally honest. The closure verification multi-week burden is appropriately tier-gated; the team’s self-correction discipline (v0.5 → v0.6) is exemplary; the closure-condition matrix C1-C7 gives Elie a clean computational target and Lyra a clean structural target for v0.7+.

— Keeper. K3 cluster K-audit CONDITIONAL PASS; 6 closure conditions for v0.7+; multi-week roadmap sequenced; cross-CI handoffs specified; Saturday’s substantive bulk-color advance ratified at CANDIDATE COUNT MATCH at Cartan-Weyl basis alignment.